SELF-DECLARED / PERSONAL RECORD

Data Science Project Portfolio

Nusrat Tabassum Nisa

Prepared 21 October 2026 | github.com/nisa3131

Self-declared personal project record. This document does not certify internship completion or imply company endorsement.

Python | pandas | NumPy | scikit-learn | Matplotlib | Data analysis | Model evaluation

01 Iris Flower Classification

150 observations; logistic regression; group-aware 120/30 train/test split.

Held-out accuracy: 100%; macro F1: 1.0000 on 30 test observations.

A small educational dataset and one split; no external validation.

github.com/nisa3131/CodeAlpha_IrisFlowerClassification

02 Unemployment Analysis with Python

740 historical and 267 January-October 2020 observations, analyzed separately.

Same 25 states: 9.20% (Jan-Mar), 23.20% (Apr-May), 8.90% (Aug-Oct).

Equal-state descriptive means; not official national estimates or causal effects.

github.com/nisa3131/CodeAlpha_UnemploymentAnalysis

03 Used-Vehicle Price Prediction

299 records after deduplication; random forest; vehicle-name groups kept apart.

70-row test set: MAE 2.0562, RMSE 3.3528, R-squared 0.5839.

Historical cars, motorcycles and scooters; currency and price units unverified.

github.com/nisa3131/CodeAlpha_CarPricePrediction

04 Sales Prediction

200 advertising observations; quadratic ridge; 160/40 train/test split.

Held-out MAE 0.485, RMSE 0.621, R-squared 0.988.

Observational prediction; no dates, verified units or causal return-on-investment claim.

github.com/nisa3131/CodeAlpha_SalesPrediction

Evidence and reproducibility

Each repository contains source code, an executed notebook, analysis outputs, dependency requirements and a written report. Results above summarize the saved project analyses.